

Perot cavity and provides positive feedback of the photons by reflection at the mirrors at either end of the cavity emitting coherent light.

However, the radiative properties of a laser may be improved greatly by using heterojunctions instead of homojunctions. A heterojunction is an interface between two adjoining single-crystal semiconductors with different bandgap energies.

Today's all laser diodes have a double heterostructure wherein a thin layer of a suitable semiconductor is sandwiched between two layers of a higher-bandgap material, forming two heterojunctions. This leads to a large number of carrier recombination and photon emission at the active layer because the energy barriers provided by the heterojunctions confine the carriers to the active layer.

Another advantage of using heterojunction is that due to difference in refractive index of materials at the heterojunction, there is optical confinement. Thus all the guided light is reflected from the interface and propagated back and forth within the laser resonator formed by the active layer. The combined effect of carrier confinement, optical confinement and lower absorption loss leads to low threshold current (tens of milliamperes) and high overall efficiency of the device.

Recently, double-heterojunction lasers have been fabricated with an active layer as thin as 2 nm instead of 0.1 to 0.3 μm of conventional double-heterojunction structure. These devices are known as quantum-well lasers.

In multi-quantum-well (MQW) lasers, a number of such quantum wells are provided. In MQWs corresponding to multiple active layers, layers separating the active regions are called barriers. Carrier motion normal to the active layer in these devices is restricted, resulting in quantisation of the kinetic energy into discrete energy levels for the carriers moving in that direction. Due to this reason, the thin active layer causes drastic changes in electronic and optical properties

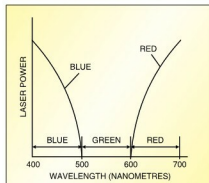


Fig. 2: Laser power for light produced in different parts of the spectrum

green part of the spectrum, the amount of power produced drops drastically (Fig. 2).

In the new approach, an exceedingly smooth, nanometres-thin layer of indium gallium nitride (InGaN) is sandwiched between two layers of GaN, forming the so-called heterostructure or quantum well (Fig. 3(b)). By applying a suitable voltage, an electric field is set up perpendicular to these layers. The electric field drives electrons and holes—positively charged particles

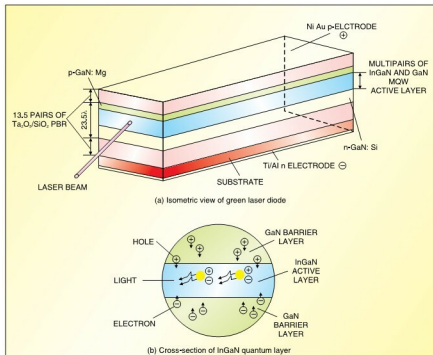


Fig. 3: MQW green laser diode

in comparison with a conventional double-heterojunction laser.

MQW devices score over conventional double-heterojunction lasers due to their lower threshold currents, narrower line-widths, higher modulation speeds, lower frequency chirps and lesser temperature dependence.

Green gap problem

Scientists have long been able to build semiconductor lasers that produce light in red parts of the spectrum. In the last decade, they conquered the blue and violet sections as well. As they try to push these lasers in the

that are devoid of electrons—together within the InGaN layers. Inside the narrow central active layers, the electrons and holes recombine, annihilating one another and generating photons with an energy precisely determined by the properties of the active semiconductor material (Fig. 3(b)). By increasing the indium concentration in the alloy, one can lower this energy, thereby increasing the wavelength of the light and changing its colour from violet to blue to green.

But, there is a flaw—the more the indium in these active layers, the more the likelihood of it pooling into small